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C23-EC/ECII-405

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BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DECE – FOURTH SEMESTER EXAMINATION

DIGITAL LOGIC DESIGN THROUGH VERILOG HDL

Time : 3 hours] [*Total Marks* : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Write about identifiers and keywords in Verilog HDL.

2. List any three advantages of hierarchical modeling.

3. Write the instantiations of AND gate and BUF gates.

4. Define rise, fall and turn-off delays in gate level modelling.

5. List any three different timing controls in behavioural modelling.

6. Write any three differences between conditional and case statements.

7. Give the three differences between various modeling in Verilog.

8. Write Verilog code for 4-to-1 multiplexer using behavioural modeling.

9. List four important programmable logic devices.

10. List any three applications of programmable logic devices.

PART—B

10×5=50

Instructions : (1) Answer *any* **five** questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. *(a)* Compare VHDL and Verilog HDL. 5

(b) List the features of Verilog HDL. 5

12. Explain briefly about below given data types with examples *(i)* nets, *(ii)* registers and *(iii)* vectors. 10

13. Design and write Verilog code for 3 to 8 decoder using dataflow modelling. 10

14. Explain while, for looping statements with examples. 10

15. Explain sequential user defined primitives with example. 10

16. Design JK flip-flop in behavioral modeling. 10

17. Design asynchronous counter in behavioural modeling. 10

18. Explain the architecture of FPGA. 10

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